



Fig. 5: IoT web page

the Arduino sketch obtained from electronicsforu.com. Click the Upload button (right arrow icon) in the Arduino IDE. Wait for the IDE to compile and upload the code to the ESP32. You will see a message that indicates “Upload complete” once the process finishes. Use this source code for uploading into the ESP32 module.

2. Install the necessary libraries.

- Install Arduino IDE
- Download the necessary libraries from the following links:
 - <https://github.com/me-no-dev/AsyncTCP>
 - <https://github.com/me-no-dev/ESPAsyncWebServer>
 - <https://github.com/jkb-git/ESP32Servo>

• Add the Library to Arduino IDE:

- Open the Arduino IDE
- Go to Sketch→Include Library→Add .ZIP Library.
- Select the .zip file and click OK.
- The library will now be installed and available in the Arduino IDE.

3. Upload source code into ESP32.

- Open Arduino IDE.
- Go to File→Preferences.
- In the Additional Boards Manager URLs field, add:

https://dl.espressif.com/dl/package_esp32_index.json
- Click OK.
- Go to Tools→Board→Boards manager, search for ESP32, and install it.
- Select the correct ESP32 board model under Tools→Board (e.g.,

ESP32 dev module).

- Under Tools→Port, select the corresponding COM port for your ESP32 device.

- Upload the saved source code into the ESP32.

4. Connect the ESP32 to the ESC for PWM control. After uploading the source code, connect GPIO 15 of the ESP32 to the PWM input pin on the ESC (orange). This will send the PWM signal to control the motor speed. You will use the ESP32Servo.h library to generate a 50Hz PWM signal on GPIO 15. The duty cycle will control the motor speed: higher duty cycles increase the motor speed while lower duty cycles decrease it.

5. Connect input wires to ESC. Connect the three-phase wires of the BLDC motor to the ESC’s motor terminals. These wires will carry the signals that control the motor’s rotation and power delivery.

6. Power the ESP32 using 5V from ESC. Connect the 5V (red) and GND (brown) pins from the ESC to

the 5V and GND pins on the ESP32. This connection will power the ESP32 directly from the ESC’s 5V output, eliminating the need for an external power supply.

7. Power the ESC. Connect the 12V power supply to the ESC’s power input. The 12V power supply should meet the voltage and current requirements for the ESC and BLDC motors. The ESC will regulate the motor’s power based on the PWM signal.

8. Turn on power. Power up the ESC by connecting the 12V power supply. The BLDC motor will beep to indicate that the ESC has initialised and is ready for PWM control.

Testing

1. Connect to the ESP32 network.

Once the ESP32 is powered up, it will create a Wi-Fi network with the SSID “DIY_BLDC” Password “987654321”. Connect your smartphone, tablet, or laptop to this Wi-Fi network.

2. Access the web interface.

Open a web browser and go to <http://192.168.4.1> (the IP address of the ESP32 in AP mode). This will load the web page (see Fig. 5) with the motor speed control slider. Fig. 5 shows the IoT web page.

3. Control the motor speed. Use the slider on the web page to adjust the motor speed. The motor speed will change according to the PWM signal sent to the ESC. **EFY**

EFY Note
The source code
of this project
is available
for free download at
www.electronicsforu.com



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